

Question bank: what we have, what's missing, how it flows

Findings from reading all 305 papers in the question-bank repo, the schema changes we need, and how questions get from a PDF to a student's screen.

What we have today

PAPERS 305 All parsed cleanly	QUESTIONS 18,651 Real, usable volume
OPTIONS PARSED 97.8% 419 have none — broken	HAVE AN ANSWER 0% Not one

The good news: **the filters we want already work**. Bank, role, exam type and year are filled in on 93–97% of papers. A student can filter to "SBI, Mains, 2019" today.

The blocker: no question has an answer

answer	0 / 18,651	0.0%
explanation	0 / 18,651	0.0%
topic	0 / 18,651	0.0%
section	2,890 / 18,651	15.5%

The fields exist. They are all empty. We can *show* a student a question but we cannot tell them if they got it right.

Without answers there is no marking, no score, no mastery, no weak-area practice, and nothing for the grade rail to measure. It is a PDF viewer, not a question bank. **This is a data problem, not a code problem** — the parser has to pull answer keys out, or someone keys them in.

Why paper_id has to stay

This came up as a question. The data settles it.

```
distinct (paper_id, direction_id) pairs : 3,039
distinct direction_ids                  :    30
reused across papers                   :    30 of 30 ← every one
```

Passages are numbered `d001 ... d030` *inside each paper*, so the same ID appears in dozens of papers. 71% of our questions point at one.

Drop `paper_id` and we show the wrong passage with the wrong question. A student reads a comprehension passage that has nothing to do with the blanks they're filling in. That isn't messy data, it's a broken question on screen.

Schema changes

Nothing needs removing. Four fields need filling, six need adding.

FIELD		WHY
answer	FILL	Exists but empty. Nothing works without it.
explanation	FILL	The learning moment. Half of what ₹200 buys.
topic	FILL	Weak-area practice and mastery need it.
section	FILL	Only 15% labelled — can't filter by section yet.
marks negative_marks	ADD	Banking exams have negative marking. A mock cannot be scored without this.
difficulty	ADD	Adaptive practice and balanced mock papers.
content_hash	ADD	Same question from two coaching centres = one entry.
exam_key	ADD	Groups the several papers that recall one real exam, so a student sees one mock, not three near-copies.
is_active	ADD	Retire a bad question without deleting its history.
section_duration	ADD	Prelims are sectionally timed. Needed for real mocks.
paper_id	KEEP	See above — passages break without it.
direction_id	SCOPE	Only unique <i>within</i> a paper. Always pair with <code>paper_id</code> .
options	TYPE	Currently untyped — 419 questions have zero options and nothing rejects them.

How the flow works

Four stages, in order. Each one is independent — we can build and test them one at a time.

01 PDF → JSON

Already done. The pipeline reads the PDFs and writes one JSON file per paper, with the questions nested inside. 305 files, all clean.

02 JSON → database

A script reads those files into two Postgres tables. Papers stay papers; questions become rows we can query. Duplicates are caught by `content_hash`.

```
papers      paper_id · exam_key · bank · role · year · shift
questions   q_id · paper_id · section · topic · answer · options
```

03 Filters live in the web address

Not folders. Folders force one order — bank, then role, then year — and break the moment someone wants "all Reasoning from 2019, any bank". Putting filters in the address means every combination works, links can be shared, the back button behaves, and Google can index each one.

```
/question-bank?bank=SBI&year=2019&section=Reasoning
```

04 Server reads the address and sends finished HTML

The server turns those filters into one database query and sends the page with the questions already in it. No loading spinner, and search engines see the questions.

One table, two views

The same rows serve both screens. Only the grouping differs.

SCREEN	HOW IT READS THE DATA	WHAT THE STUDENT DOES
Question bank	filter by bank / year / section	"Give me 20 Reasoning questions from SBI 2019"
Mock test	group by paper_id, order by q_num	"Sit the whole SBI PO Mains 2019 paper, timed"

Filters power the first. `paper_id` powers the second, and keeps passages attached to the right questions in both.

What to do first

- **Get answers into the data.** Everything else is blocked behind this. Nothing we build has value until a student can be told they're wrong.
- **Load the 305 files into Postgres** with the new columns. Can start now — doesn't wait on answers.
- **Ship the bank with bank / role / year filters.** Works with today's data.
- **Classify sections and topics** for the 84.5% that are unlabelled. That unlocks topic filters, mastery, and weak-area practice.

Figures from reading every file under `India/banking/papers-deduped/` at commit `0ff28ea` . Percentages are of 18,651 questions across 305 papers.